## 1. Architecture: Airflow + MLflow Interaction

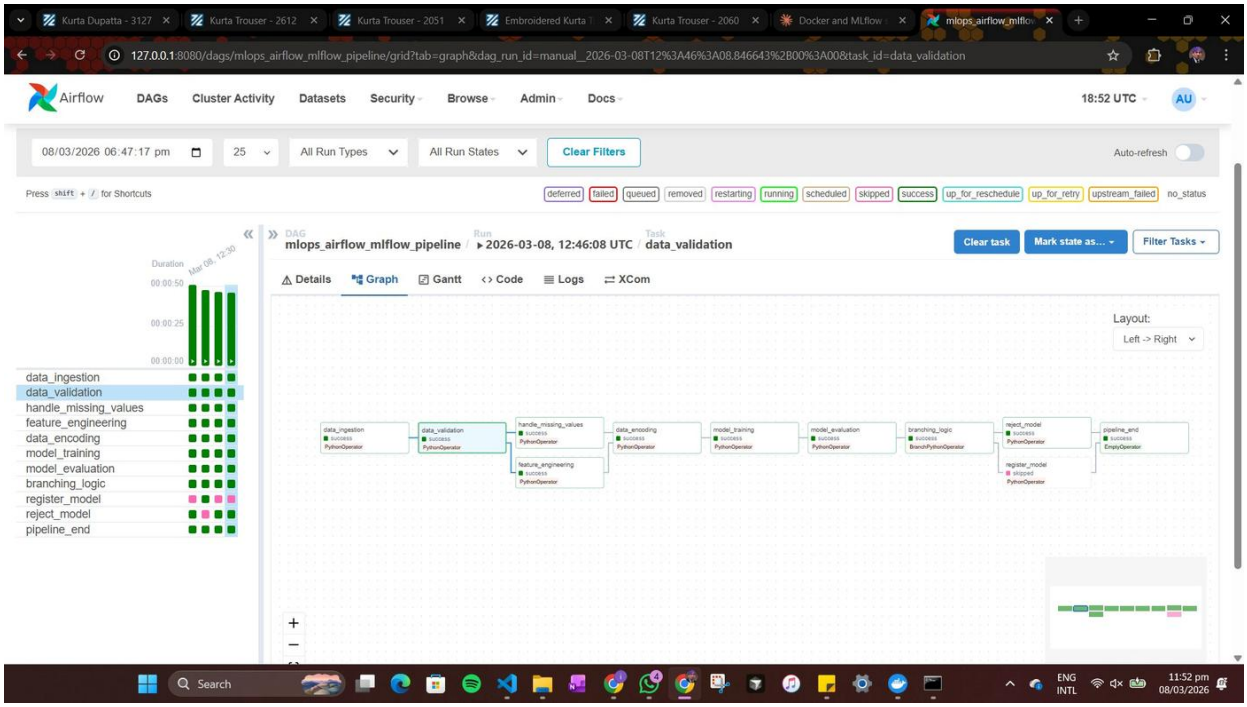


*Figure 1: Airflow DAG Graph View — Full Pipeline*

## 2. DAG Structure and Task Dependencies

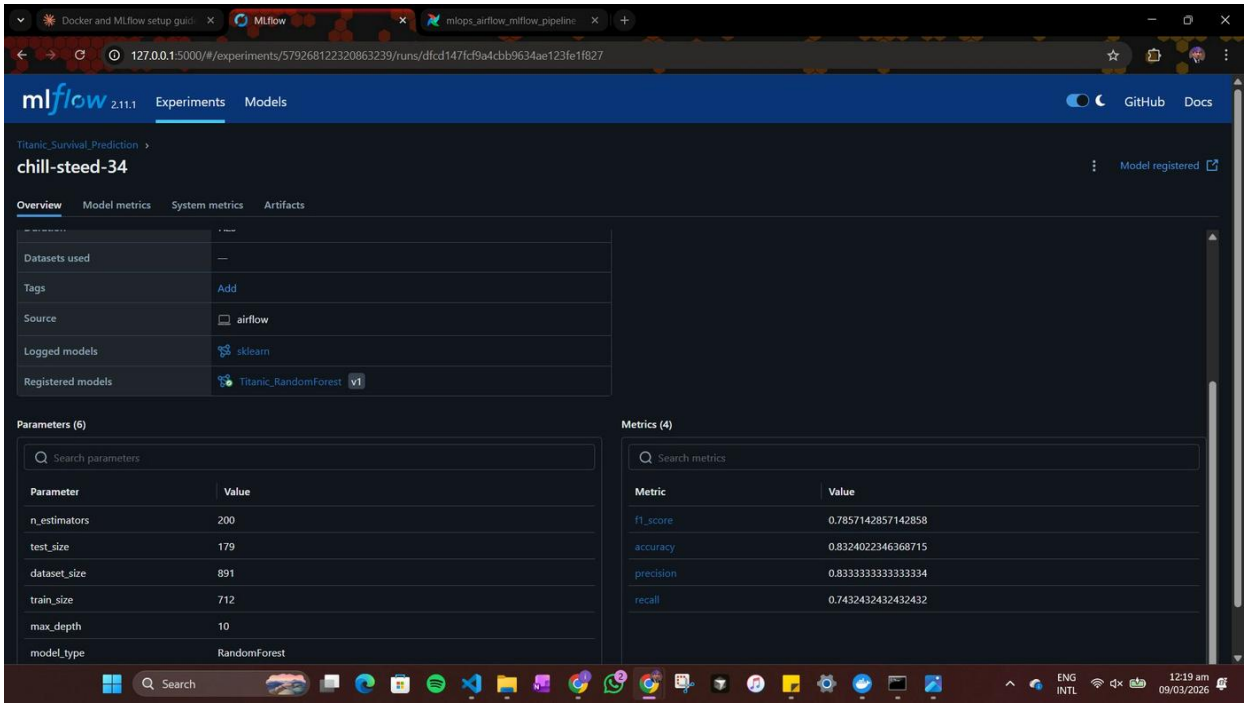


*Figure 2: Single Run Detail — chill-steed-34 (RF n=200, d=10, Accuracy: 0.832)*

## 3. Failure and Retry Explanation

The data_validation task is configured with retries=2 and retry_delay=5 seconds. An intentional ValueError is raised on attempt 1 to simulate a transient failure and demonstrate Airflow's retry mechanism.

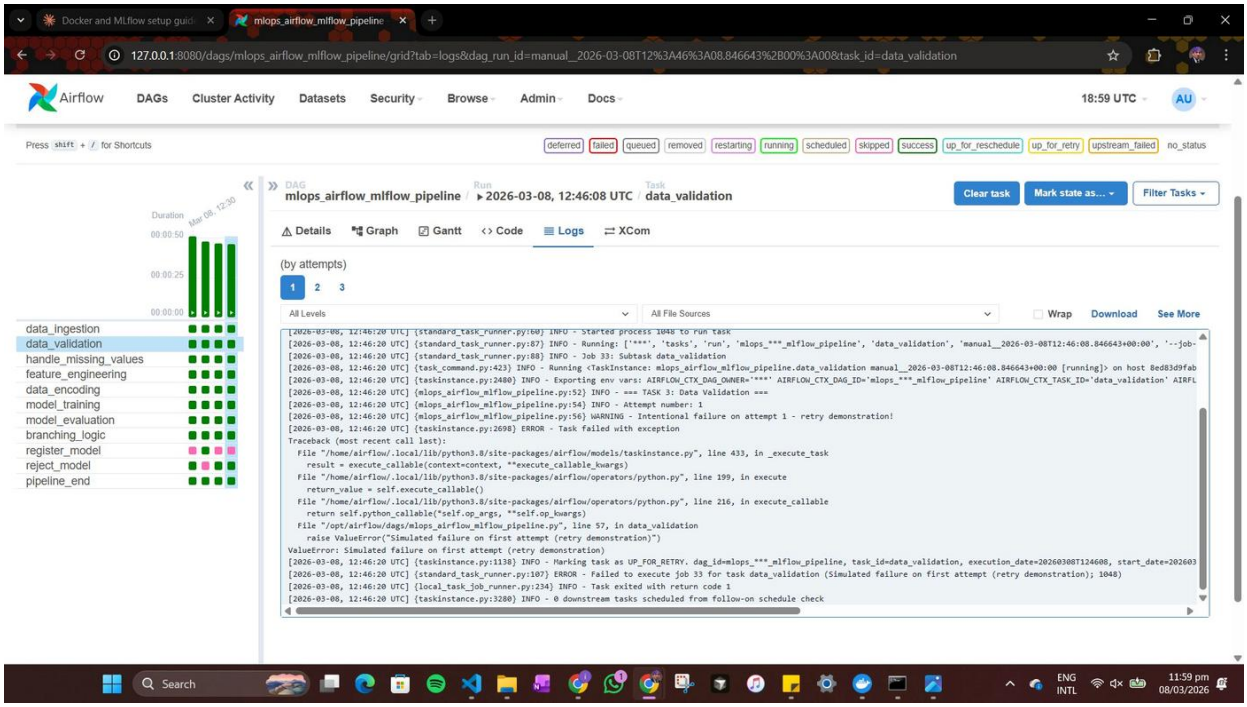


*Figure 3: Attempt 1 — ValueError raised, task marked UP_FOR_RETRY*

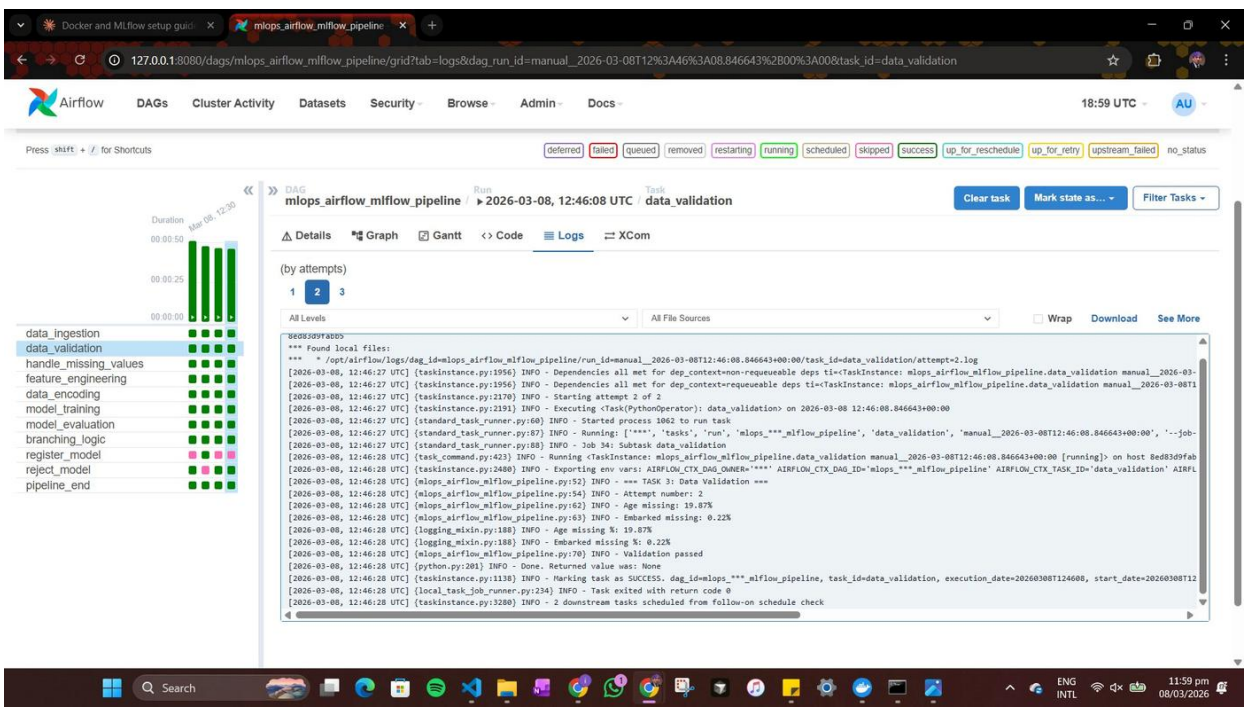


*Figure 4: Attempt 2 — Validation passed, Age missing 19.87%, Embarked 0.22%*

## 4. Experiment Comparison Analysis

The DAG was executed 4 times with varying hyperparameters. All runs are tracked in MLflow under the Titanic_Survival_Prediction experiment. Results are compared below:

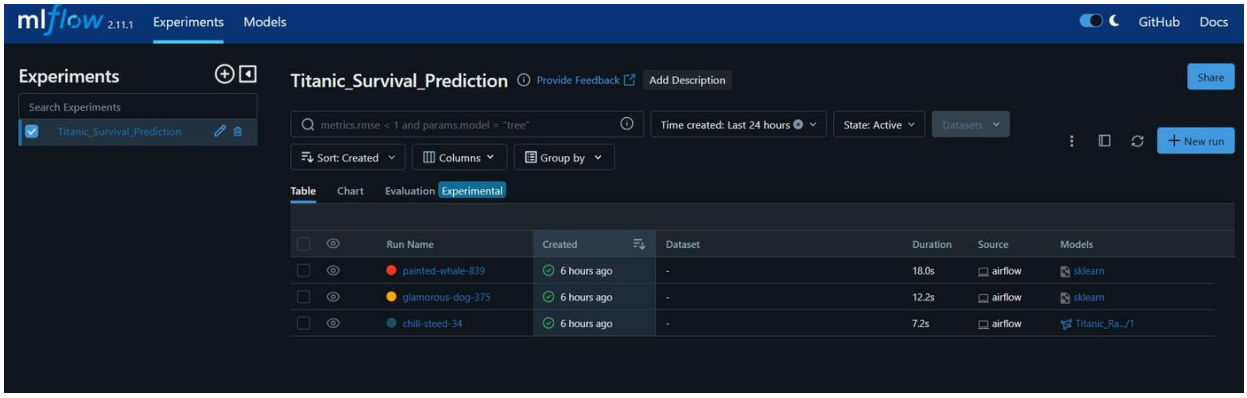


*Figure 5: MLflow Experiments Page — All Runs Listed*

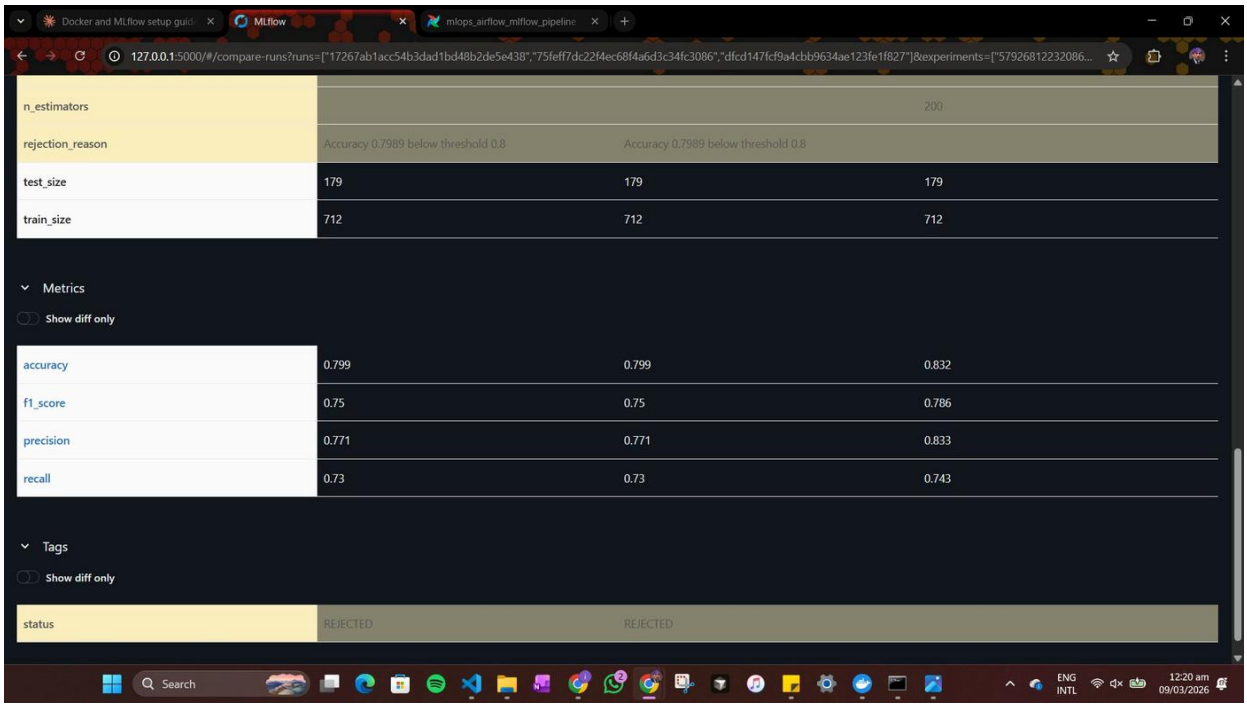


*Figure 6: MLflow Run Comparison — Metrics Table (accuracy, f1, precision, recall)*

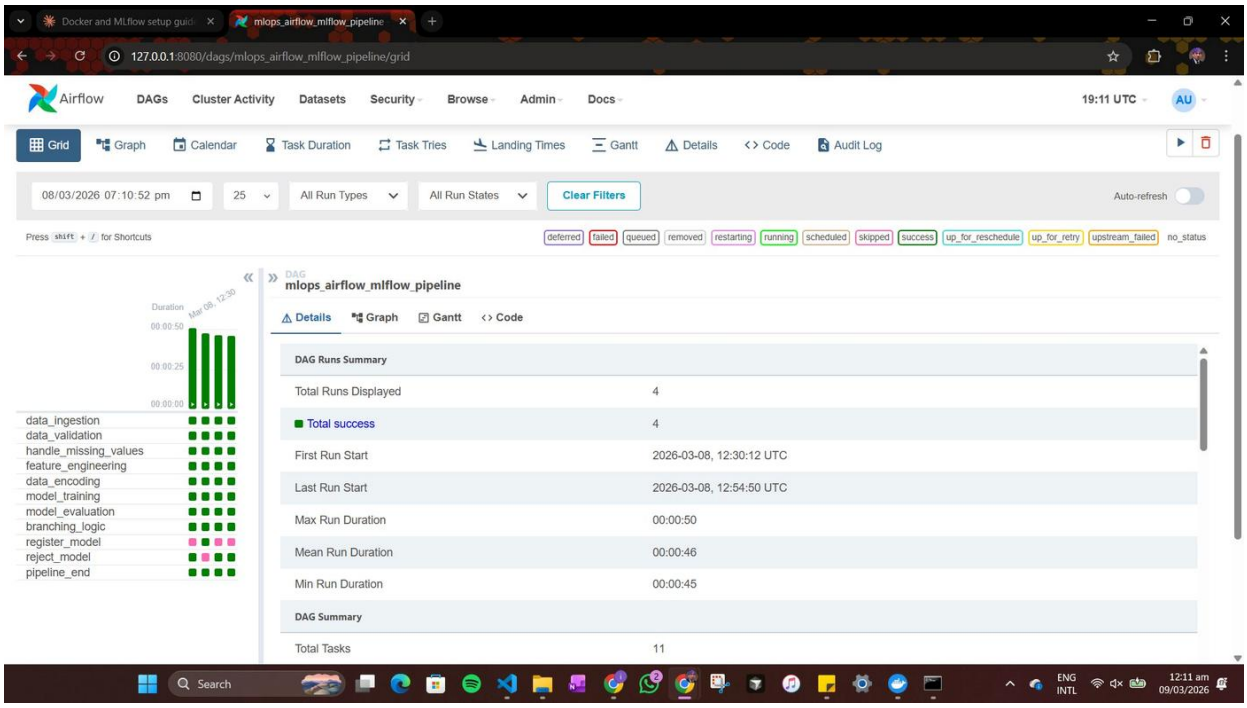


*Figure 7: Airflow — All 4 DAG Runs Summary (Total success: 4)*

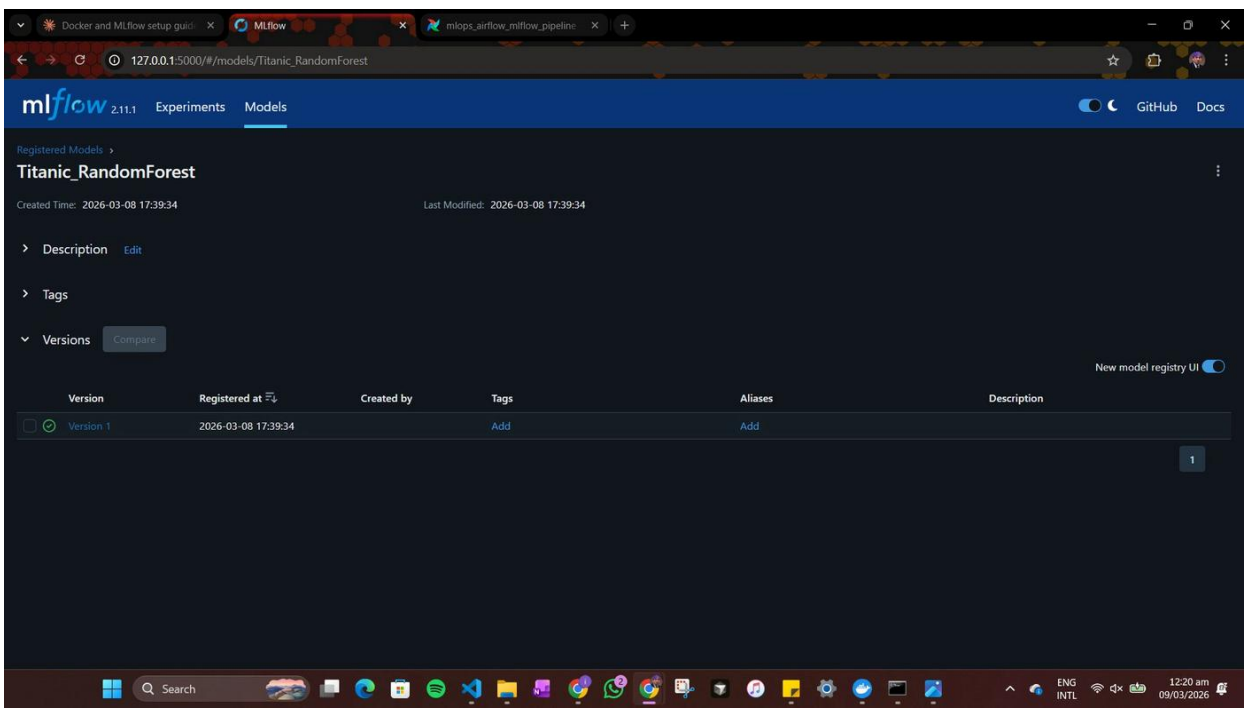


*Figure 8: Titanic_RandomForest — Version 1 Details*

**Best Model: RandomForest with n_estimators=200, max_depth=10 (Run: chill-steed-34)**

This run achieved the highest accuracy (83.2%), F1-score (0.786), and precision (0.833). Increasing tree count and depth allowed the model to capture more complex survival patterns. LogisticRegression performed worst (73.9%) because it assumes linear boundaries, which cannot fully model the interaction effects (e.g., class × gender) present in the Titanic dataset.

## 5. Model Registration

When accuracy meets the threshold, the register_model task calls mlflow.register_model() to add the trained model to the MLflow Model Registry. Runs below the threshold have their rejection reason logged instead.

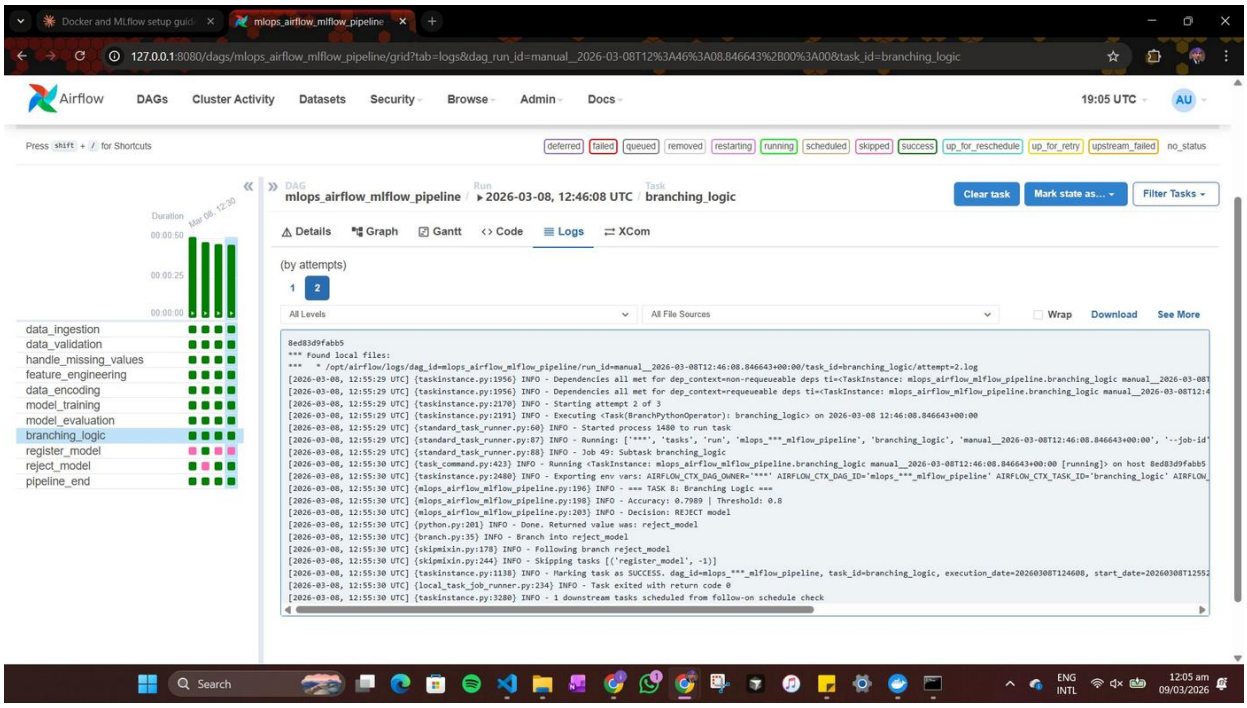


*Figure 9: Branching Logic — REJECT decision (Accuracy 0.799 < threshold 0.8)*

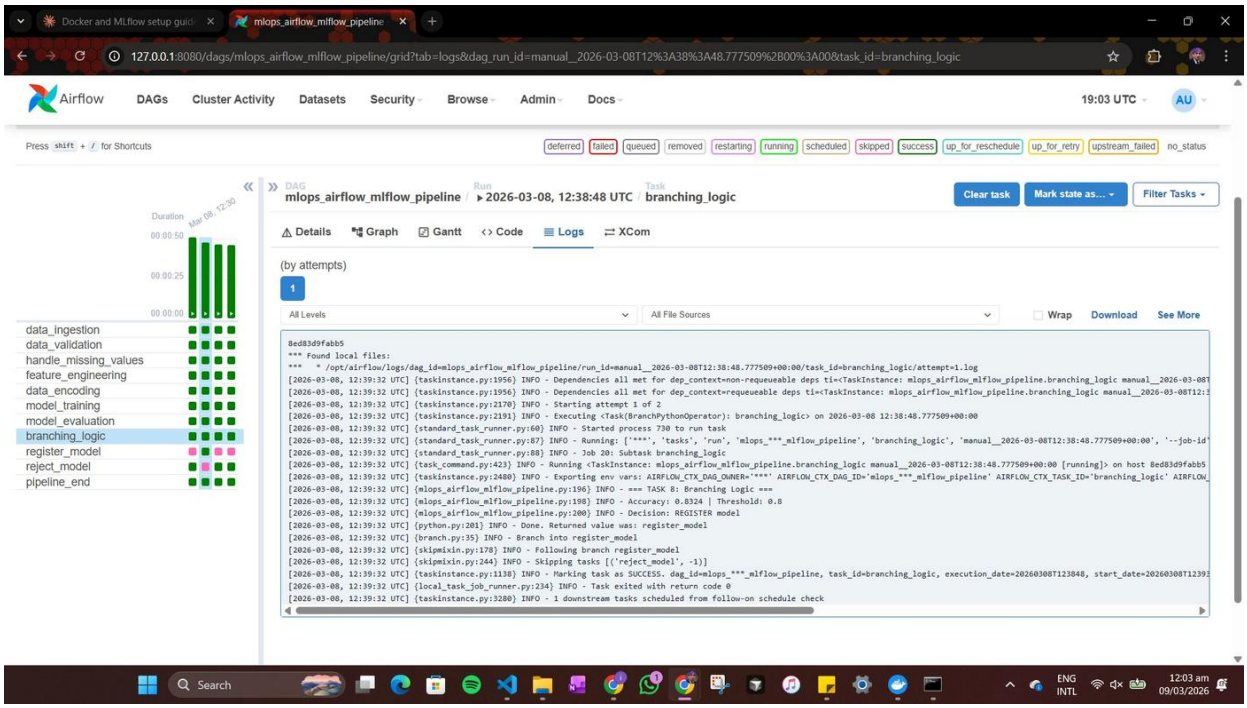


*Figure 10: Branching Logic — REGISTER decision (Accuracy 0.832 >= threshold 0.8)*

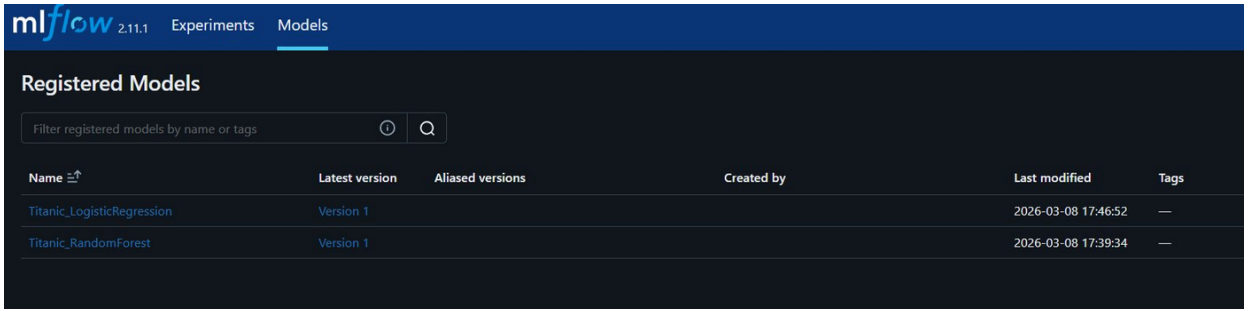


*Figure 11: MLflow Model Registry — Both Titanic_RandomForest and Titanic_LogisticRegression registered*

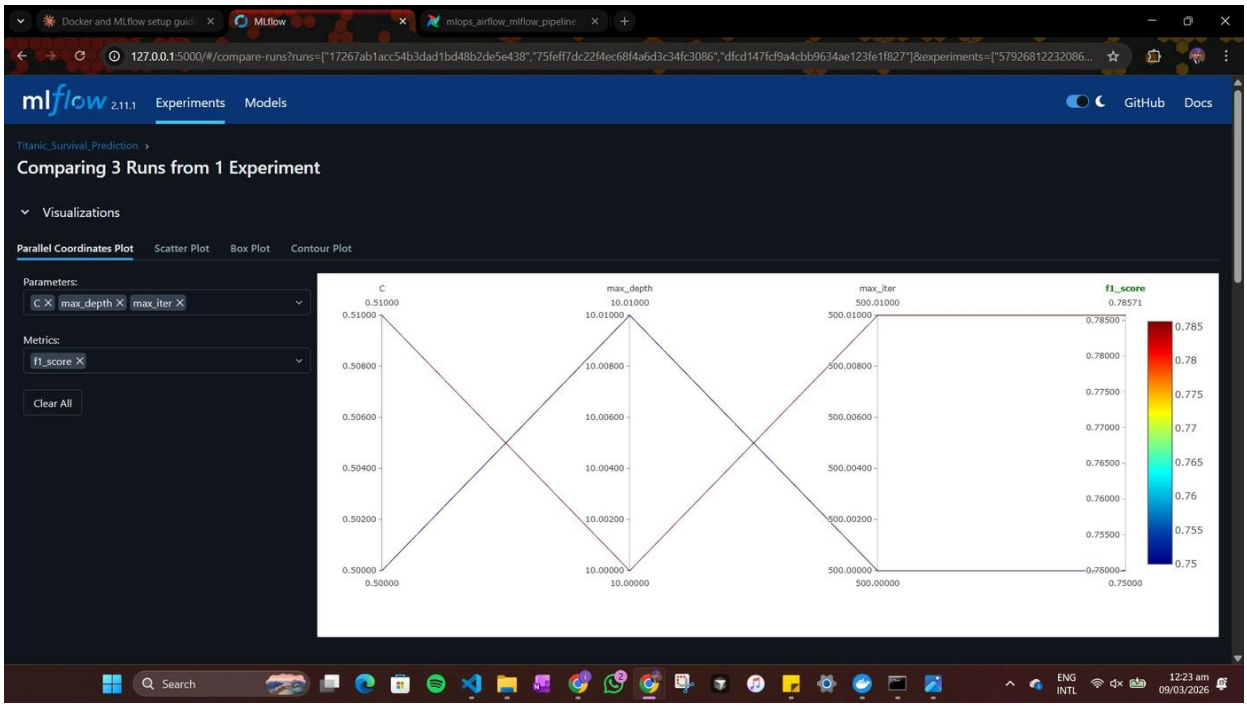


*Figure 12: Comparison Chart- Parallel coordinates plot*